Physics A (H156, H556)

Materials
KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: Not set

Candidate forename		Candidate surname	

Centre number						Candidate number				

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

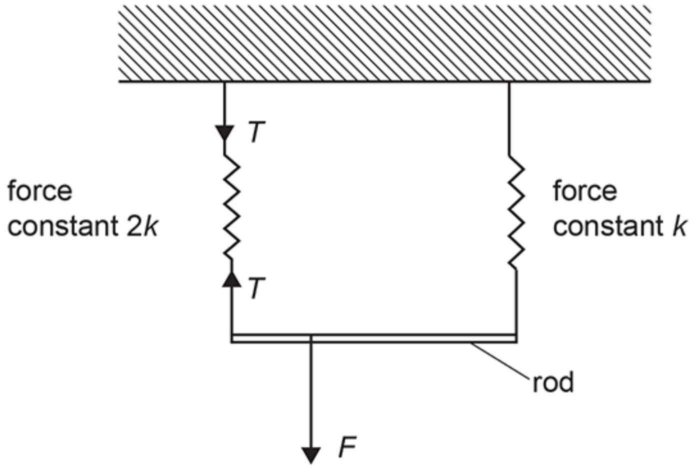
INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **82**.
- The total number of marks may take into account some 'either/or' question choices.

1 A light rod is hung from a horizontal ceiling using 2 springs, one attached to each end. The springs have force constants k and $2k$.

A downwards force F is applied closer to one end of the rod such that the rod is horizontal.

What is the tension T in the spring with the force constant $2k$?



A $\frac{F}{3}$

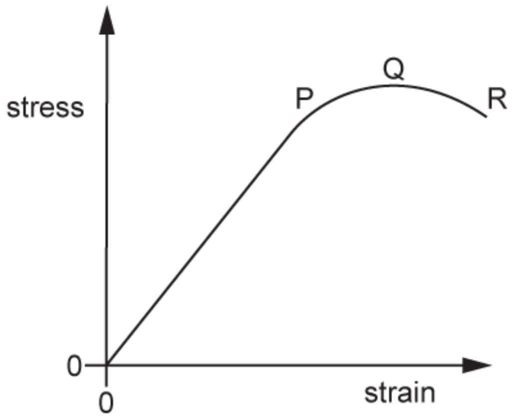
B $\frac{F}{2}$

C $\frac{2F}{3}$

D F

Your answer [1]

2 Which row in the table correctly identifies the elastic limit, fracture and ultimate tensile strength in the graph below?



	Elastic limit	Fracture	Ultimate tensile strength
A	P	Q	R
B	P	R	Q
C	P	R	R
D	Q	R	P

Your answer **[1]**

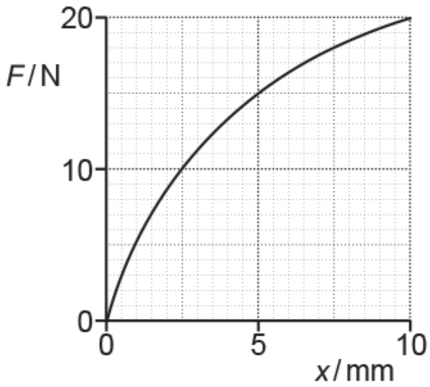
3 A wire of cross-sectional area 3.9×10^{-6} m^2 carries a load of 240 N. The strain in the wire is 0.30%.

Which value of the Young modulus, in Pa, is correct and expressed to an appropriate number of significant figures?

A 2.05×10^8
B 2.1×10^8
C 2.05×10^{10}
D 2.1×10^{10}

Your answer **[1]**

4 The force F against extension x graph for a material being stretched is shown.



What is best estimate for the energy stored in the material when the extension is 10 mm?

A 0.07 J

B 0.10 J

C 0.13 J

D 0.20 J

Your answer **[1]**

5 A spring is stretched by hanging on it a variable mass m. The mass m is always at rest. The spring obeys Hooke's law.

What is the relationship between the elastic potential energy E in the spring and the mass m?

A $E \propto m^{-1}$

B $E \propto m^{-2}$

C $E \propto m$

D $E \propto m^{2}$

Your answer **[1]**

6 The Young modulus E of a metal can be determined using the expression $E = \frac{4F}{\varepsilon \pi d^2}$, where F is the tension in the wire, d is the diameter of the wire and ε is the strain of the wire.

Here is some data.

Quantity	Percentage uncertainty
F	5.3
ε	1.2
D	1.0

What is the percentage uncertainty in the calculated value of E?

A 2.1 %
B 6.4 %
C 7.5 %
D 8.5 %

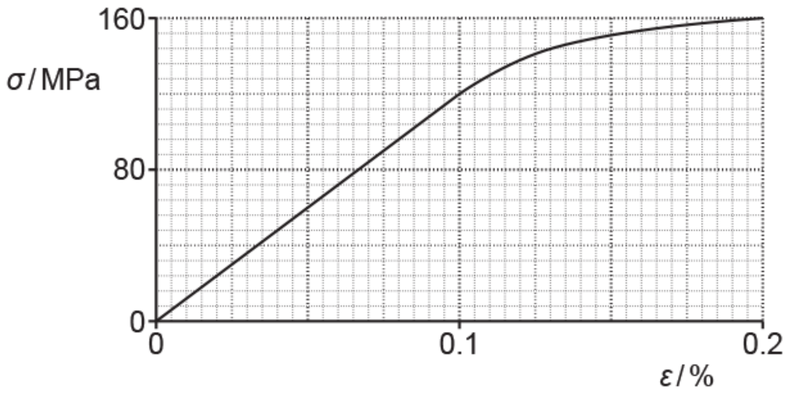
Your answer [1]

7 Which pair of quantities have the same S.I. base units?

A force, strain
B force, stress
C pressure, stress
D strain, upthrust

Your answer [1]

8 A graph showing the variation of the stress σ with strain ε for a material is shown below.



What is the Young modulus of the material?

A 6.0×10^4 Pa

B 1.2×10^9 Pa

C 8.0×10^{10} Pa

D 1.2×10^{11} Pa

Your answer [ ] **[1]**

9 The table below shows some data on two wires X and Y.

Wire	Young modulus of material / GPa	Cross–sectional area of wire / mm^2
X	120	1.0
Y	200	2.0

The wires **X** and **Y** have the same original length. The tension in each wire is the same.
Both wires obey Hooke's law.

What is the value of the ratio $\frac{\text{extension of } \mathbf{X}}{\text{extension of } \mathbf{Y}}$?

A 0.30
B 1.7
C 2.0
D 3.3

Your answer [1]

10 One end of a spring is fixed and a force F is applied to its other end. The elastic potential energy in the extended spring is *E*. The spring obeys Hooke's law.

What is the extension *x* of the spring?

A $x = \frac{E}{F}$

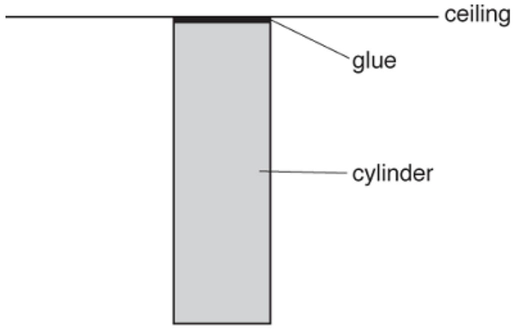
B $x = \frac{F}{E}$

C $x = \frac{2E}{F}$

D $x = \frac{F}{2E}$

Your answer [1]

11 The flat end of a uniform steel cylinder of weight 7.8 N is glued to a horizontal ceiling. The cylinder hangs vertically. The breaking stress for the glue is 130 kPa.



The glue only just holds the cylinder to the ceiling.

What is the cross-sectional area of the cylinder?

A $6.0 \times 10^{-2}\ m^2$

B $6.0 \times 10^{-5}\ m^2$

C $1.7 \times 10^{-2}\ m^2$

D $1.7 \times 10^{1}\ m^2$

Your answer [1]

12 A tensile force of 4.5 N is applied to a spring. The spring extends elastically by 3.2 cm.

What is the elastic potential energy of the spring?

A 0.072 J

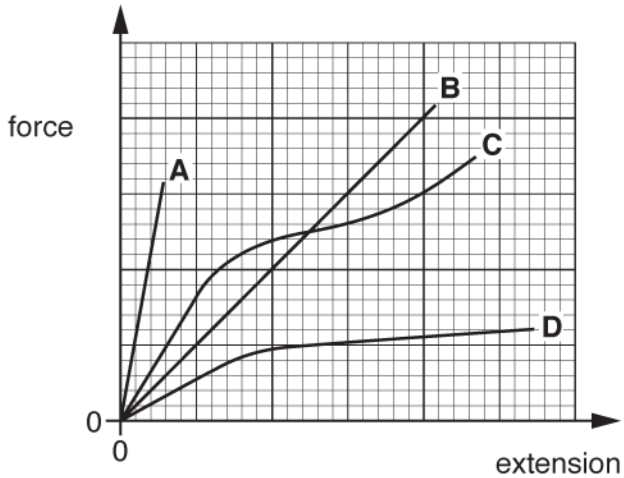
B 0.14 J

C 2.4 J

D 14 J

Your answer [1]

13 Four materials **A**, **B**, **C and D** have the same length and cross-sectional area. The force against extension graph for each material up to the breaking point is shown below.



Which material is brittle and has the greatest ultimate tensile strength?

Your answer [1]

14 A student is investigating two wires of different diameters made from the **same** metal. She plots graphs of force against extension and graphs of stress against strain for both wires.

The wires behave elastically.

Which of the following statements is / are correct?

1 Young modulus is equal to the gradient of the stress against strain graph.
2 Force constant is equal to the gradient of the force against extension graph.
3 The graphs of stress against strain have different gradients.

A 1, 2 and 3
B Only 1 and 2
C Only 2
D Only 1

Your answer

[1]

15 A 6.0 N force is applied to a spring which extends vertically downwards by a distance 5.0 cm. The force is suddenly removed so that the spring flies vertically upwards. The spring has mass 9.0 g.

What is the maximum height reached by the spring?

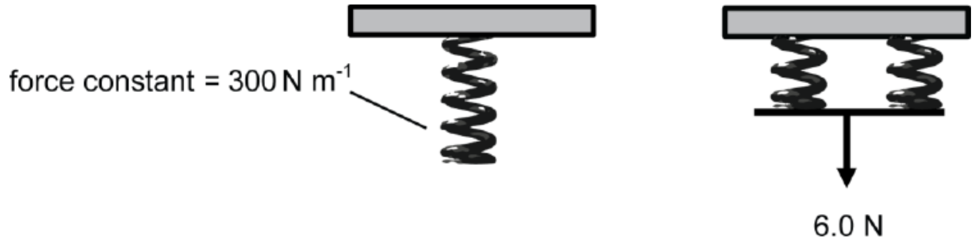
A. 0.085 m
B. 0.17 m
C. 1.7 m
D. 3.4 m

Your answer

[1]

16 A spring of force constant 300 N m^{-1} is cut in half. The two halves are then placed in **parallel** with each other. A force of 6.0 N is then applied to this parallel arrangement.

What is the extension of each spring?



A. 0.5 cm
B. 1.0 cm
C. 2.0 cm
D. 4.0 cm

Your answer

[1]

17 A wire of diameter 0.80 mm is stretched by a force of 40 N.

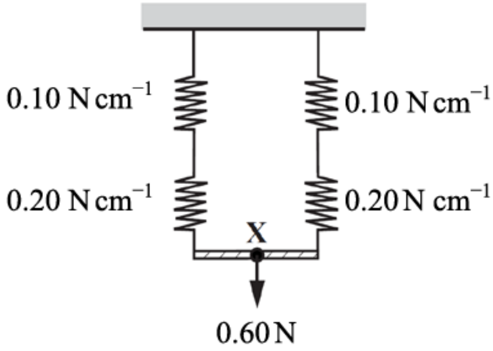
What is the tensile stress in the wire?

A. 0.016 MPa
B. 0.05 MPa
C. 20 MPa
D. 80 MPa

Your answer

[1]

18 A spring with force constant 0.10 N cm^{-1} is placed in series with one of 0.20 N cm^{-1}. These are then placed in parallel with an identical set of springs as shown. A force of 0.60 N is applied.



What distance does the point **X** move down when the 0.60 N force is applied?

A. 2.0 cm
B. 3.0 cm
C. 4.5 cm
D. 9.0 cm

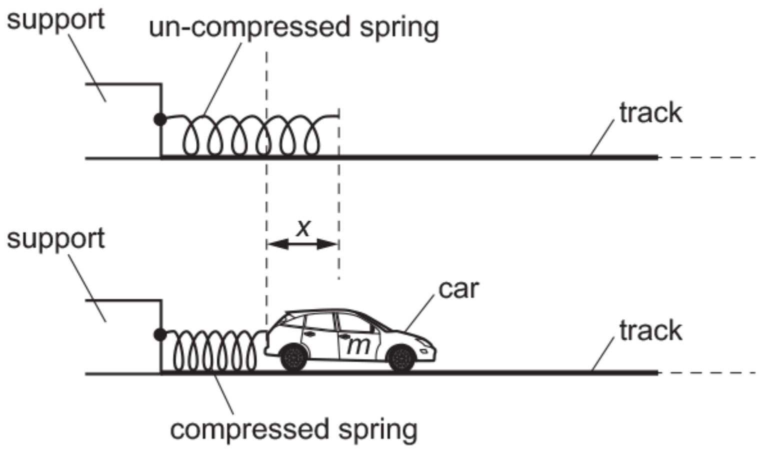
Your answer

[1]

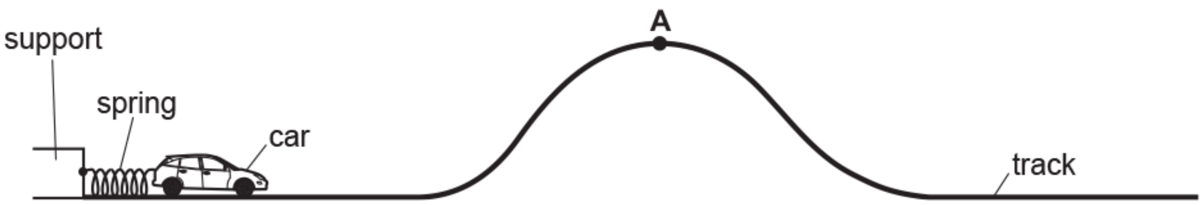
19 One end of a spring is fixed to a support.

A toy car, which is on a smooth horizontal track, is pushed against the free end of the spring.

The spring compresses. The car is then released. The car accelerates to the right until the spring returns back to its original length.



The arrangement is used to propel the toy car along a smooth track.



i. Point **A** is at the top of the track.
   The launch speed of the car is now adjusted until the car just reaches **A** with zero speed.
   The height of **A** is 0.20 m above the horizontal section of the track.

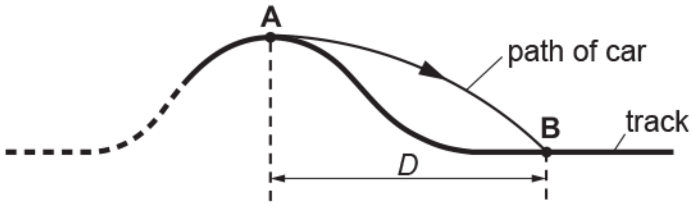
   - mass of car m = 80 g
   - force constant k of the spring = 60 N m^{-1}.

   All the elastic potential energy of the spring is transferred to gravitational potential energy of the car.

   Calculate the initial compression x of the spring.

x = ... m **[3]**

ii. At a specific speed, the car leaves point **A** horizontally and lands on the track at point **B**.
The horizontal distance between **A** and **B** is D.



Air resistance has negligible effect on the motion of the car between **A** and **B**.

1 Explain how the time of flight between **A** and **B** depends on the speed of the car at **A**.

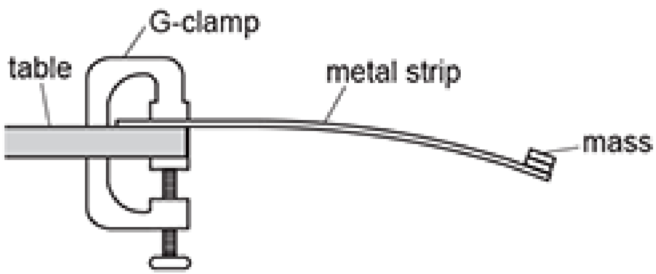
[2]

2 Explain how the distance *D* depends on the speed of the car at **A**.

[2]

20 A student wants to determine the Young modulus *E* of a metal strip.

The student clamps the metal strip to the edge of a table using a G-clamp. A mass is **permanently** fixed to the end of the strip as shown.



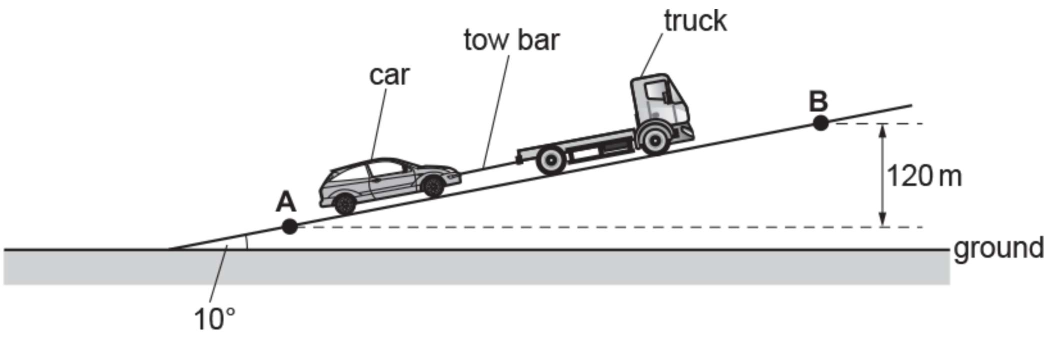
The mass oscillates freely when it is moved away from its equilibrium position and then released.

The Young modulus *E* of the metal can be determined using the equation $E = \frac{16\pi^2 mL^3}{wt^3T^2}$,where *m* is the mass fixed to the end of the strip, *L* is the length of the strip from the end of the table to the centre of the mass, *w* is the width of the strip, *t* is the thickness of the strip, and *T* is the period of oscillations.
Describe how an experiment may be safely conducted, and how the data can be analysed to determine an accurate value for *E*.

[6]

21(a) A truck pulls a car up a slope at a **constant** speed.

The truck and the car are joined with a steel tow bar, as shown in the diagram.



The diagram is **not** drawn to scale.

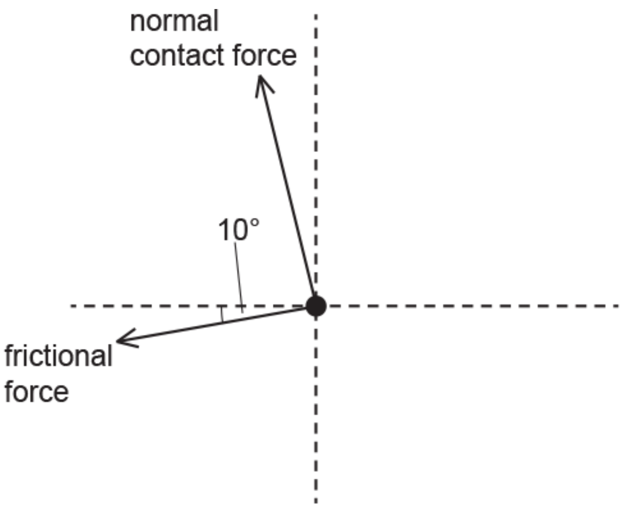
The slope is 10° to the horizontal ground.

The mass of the car is 1100 kg.

The car travels from **A** to **B** . The vertical distance between **A** and **B** is 120 m.

There are four forces acting on the **car** travelling up the slope.

Complete the free-body diagram below for the car and label the missing forces.



[2]

(b) Show that the component of the weight of the car *W*s acting down the slope is about 1900 N.

[1]

(c) The total frictional force acting on the car as it travels up the slope is 300 N.

Calculate the force provided by the tow bar on the car.

force = ... N **[1]**

(d) Calculate the work done by the force provided by the tow bar as the car travels from **A** to **B** .

work done = ... J **[3]**

(e) The steel tow bar used to pull the car has length 0.50 m and diameter 1.2×10^{-2} m.
The Young modulus of steel is 2.0×10^{11} Pa.
The force on the tow bar is 2200 N.

Calculate the extension *x* of the tow bar as the car travels up the slope.

x = ... m **[3]**

22(a) Explain what is meant by the **ultimate tensile strength** of a material.

[1]

(b) A footbridge is supported by a number of metal cables of the same length.
Each cable has uniform cross-section and diameter 4.20 mm as shown in Fig. 16.1.

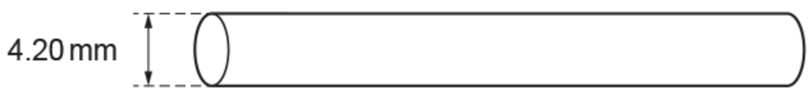


Fig. 16.1 (not to scale)

A group of engineers investigate how the extension *x* varies with applied force *F* for one of the cables.

The results of the investigation are shown in Fig. 16.2.

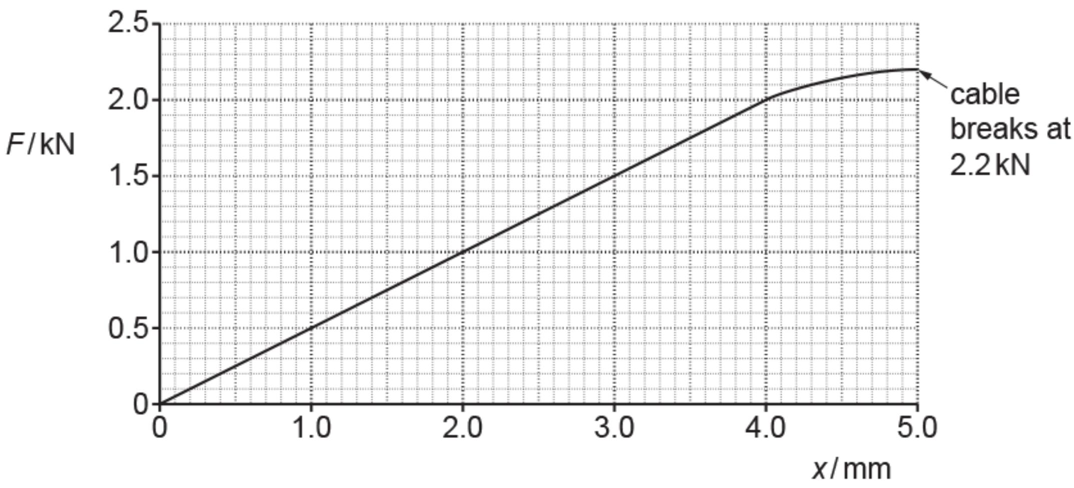


Fig. 16.2

The cable breaks when the force is 2.2 kN.

i. Describe how a suitable measuring device may have been used by the engineers to demonstrate that the cable had uniform cross-section.

---

[2]

---

ii. State any value of F when the cable behaves

1. elastically

F = ... kN

2. plastically.

F = ... kN

[2]

iii. Use Fig. 16.2 to determine the force constant k in N m^{-1} of the cable.

k = ... N m^{-1} **[2]**

(c)

Determine the breaking stress σ of the cable.
Assume that the cross-sectional area of the cable remains constant during the test.

σ = ... Pa **[2]**

(d)

Explain why the work done on the cable when its extension changes from 3.0 mm to 4.0 mm is greater than when its extension changes from 1.0 mm to 2.0 mm.

[2]

23(a) The ball-release mechanism of a pinball machine is shown in Fig. 17.1.

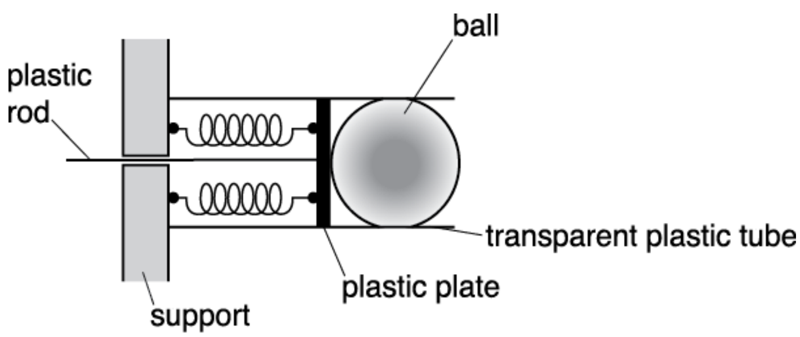


Fig. 17.1

A pair of identical compressible springs are fixed between a plastic plate and a support. The springs are in parallel. A plastic rod attached to the plate is pulled to the left to compress the springs. A ball, initially at rest, is fired when the plate is released.

A group of students are conducting an experiment to investigate the ball-release mechanism shown in Fig. 17.1. The students apply a force *F* and measure the compression *x* of the springs.
The table below shows the results.

F / N	x / cm
1.1 ± 0.2	2.0
2.0 ± 0.2	4.0
2.9 ± 0.2	6.0
4.0 ± 0.2	8.0
5.1 ± 0.2	10.0

Fig. 17.2 shows four data points from the table plotted on a F against x graph.

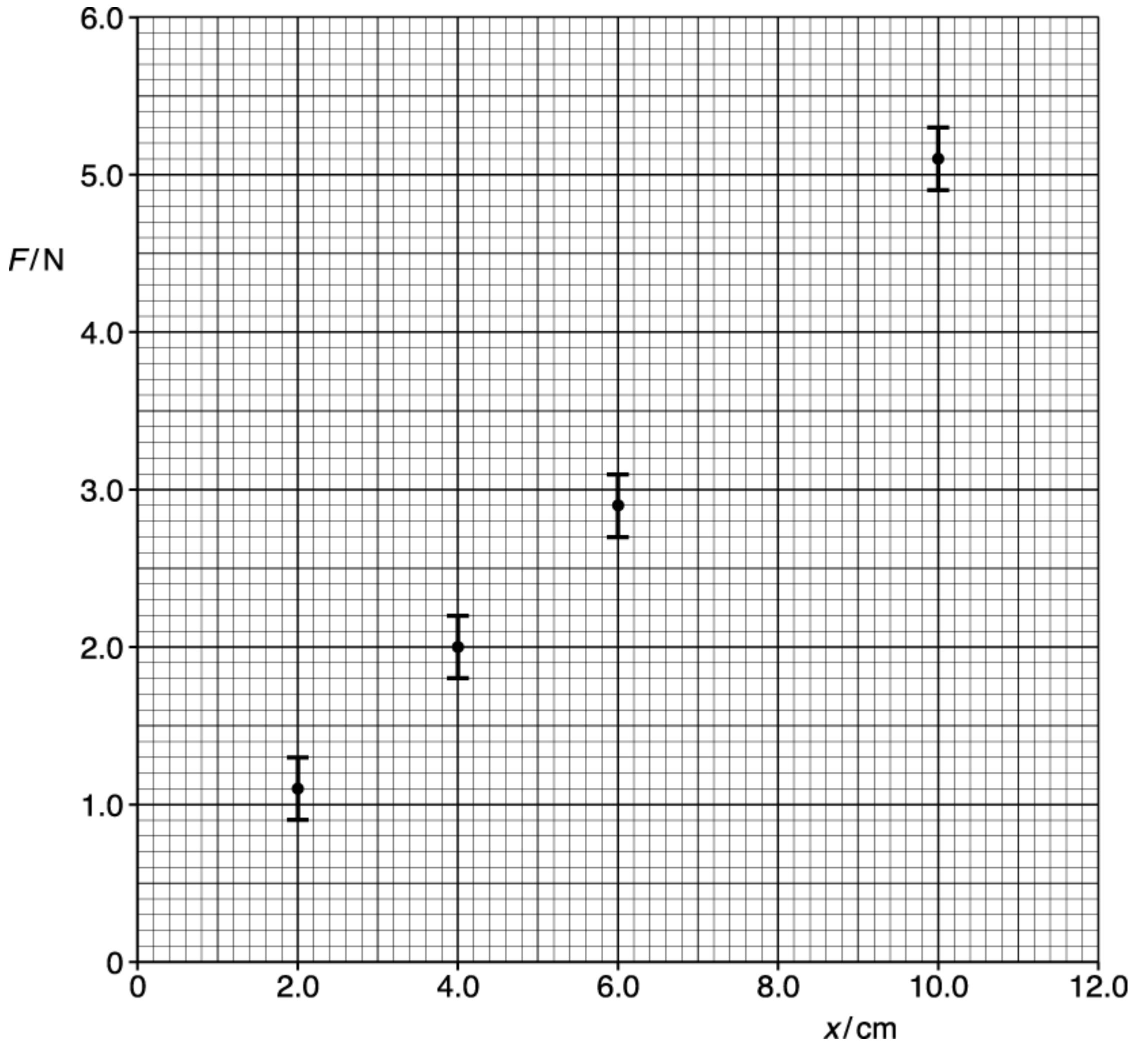


Fig. 17.2

i. Plot the missing data point and the error bar on Fig. 17.2.

[1]

ii. Describe how the data shown in the table may have been obtained in the laboratory.

-- **[2]**

iii. Draw the best fit and the worst fit straight lines on Fig. 17.2.

Use the graph to determine the force constant k for a **single** spring and the absolute uncertainty in this value.

k = ______________ ± ______________ N m^{-1} **[4]**

iv. State the feature of the graph that shows Hooke's law is obeyed by the springs.

-- **[1]**

v. The mass of the ball is 0.39 kg.

Use your answer from **(iii)** to calculate the launch speed v of the ball when the plastic plate shown in Fig. 17.1 is pulled back 12.0 cm.

v = ____________________________ m s^{-1} **[3]**

(b) A new arrangement for the ball-release mechanism using three identical springs is shown in Fig. 17.3.

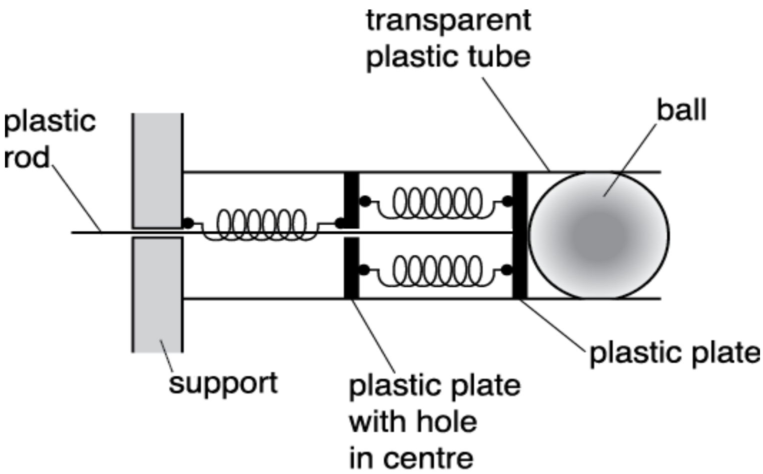


Fig. 17.3

The force constant of each spring is k.
The same ball of mass 0.39 kg is used. The plastic rod is pulled to the left by a distance of x.

Show that initial acceleration a of this ball is given by the equation

$$a = 1.7\ kx.$$

[2]

24 Fig. 18.2 shows an arrangement for lifting a car engine in a repair workshop.

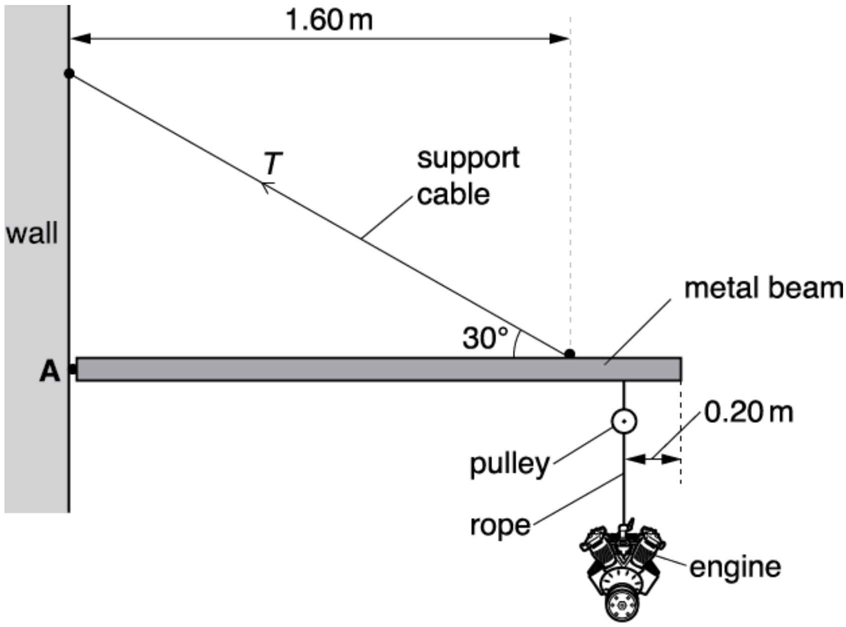


Fig. 18.2 (not to scale)

A **uniform** metal beam of length 2.00 m is hinged to a vertical wall at point **A**. The beam is held at rest in a horizontal position by a support cable of diameter of 3.0 cm. One end of this cable is fixed to the wall and the other end is fixed to the beam at a perpendicular distance of 1.60 m from the wall. The support cable makes an angle of 30° to the horizontal.

The car engine is lifted and lowered using a rope and a pulley. The pulley is fixed to the lower end of the beam at a distance of 0.20 m from the far end of the beam.

The metal beam has a mass of 120 kg and the car engine has a mass of 95 kg.

i. Calculate the tension *T* in the support cable.

T = ____________________________ N **[3]**

ii. Calculate the tensile stress σ in the support cable in kPa.

σ = ____________________________ kPa **[2]**

iii. The engine is lowered using the pulley and the rope. The engine accelerates downwards.
Explain briefly the effect this would have on the tension T in the support cable.

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[1]

25 The ball-release mechanism of a pinball machine is shown in Fig. 17.1.

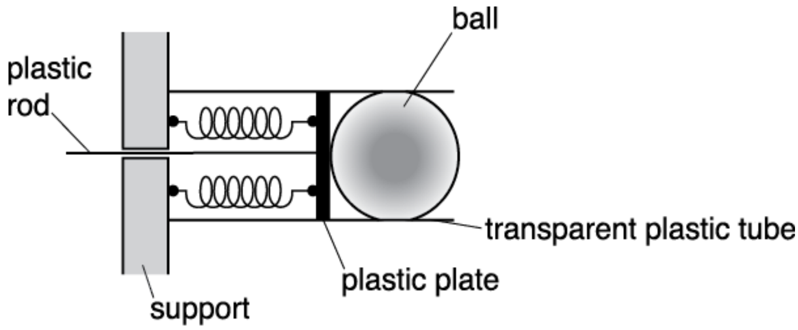

Fig. 17.1

A pair of identical compressible springs are fixed between a plastic plate and a support. The springs are in parallel. A plastic rod attached to the plate is pulled to the left to compress the springs. A ball, initially at rest, is fired when the plate is released.

The force constant of each spring is k.
Explain why the force constant of the two springs in parallel in Fig. 17.1 is equal to $2k$.

---

[1]

26(a) * A group of scientists have designed an alloy which is less dense than copper but may have similar mechanical properties. A researcher is given the task to determine the Young modulus of this alloy in the form of a wire.

Write a plan of how the researcher could do this in a laboratory to obtain accurate results. Include the equipment used and any safety precautions necessary.

[6]

(b)

i. State the meaning of *elastic* and *plastic* behaviour.

[1]

ii. Repeatedly stretching and releasing rubber warms it up.
Fig. 18.1 shows a force-extension graph for rubber.

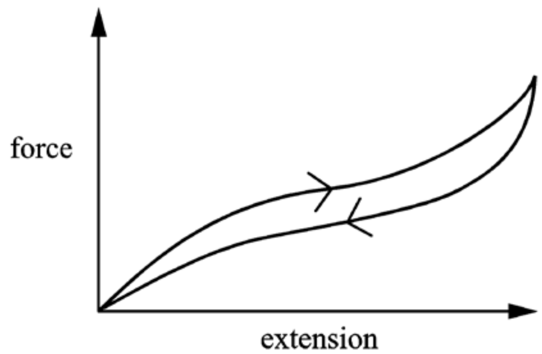

Fig. 18.1

Rubber is an ideal material for aeroplane tyres. Using the information provided, discuss the behaviour and properties of rubber and how its properties minimise the risks when aeroplanes land.

[3]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1			C	1	<u>Examiner's Comments</u> The rod ensures that the extension of both springs is the same. This means that the tension in the left spring must be twice the tension in the right spring. This in turn means that *F* must equal three times the tension of the right spring, since the sum of the two tensions must equal *F*. Finally, this gives the tension in the right spring as $\frac{F}{3}$ and hence the tension in the left spring must be $2\frac{F}{3}$.
			Total	1	
2			B	1	<u>Examiner's Comments</u> Most candidates deduced the correct answer by remembering that the ultimate tensile strength is given by the highest point on the graph, in this case, point Q.
			Total	1	
3			D	1	<u>Examiner's Comments</u> Just under half of all candidates got this correct. While the stress, given by force/area was straightforward to calculate, the strain was given as a percentage. This should have been converted to a decimal, i.e. 0.0030 to get the correct value of Young modulus.
			Total	1	
4			C	1	
			Total	1	
5			D	1	
			Total	1	
6			D	1	
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
7			C	1	
			Total	1	
8			D	1	<u>Examiner's Comments</u> The Young modulus is found by calculating the initial gradient of the material's stress-strain graph. The initial portion appears to be a straight line from the origin to the point (0.1, 120). The units on this graph are megapascals and %. This means the co-ordinates of the chosen point are in fact (0.1×10^{-2}, 120×10^6). Many candidates forgot to convert the strain into a decimal and left it as a percentage. Their answer was a factor of a 100 out, ie answer B. The correct answer is 120×10^6 / 0.1×10^2 = 1.2×10^{11} Pa. This is answer D.
			Total	1	
9			D	1	<u>Examiner's Comments</u> The tension for both wires is the same, yet wire X has half of the cross-sectional area. This means the stress for X will be twice that of Y. Strain = stress/Young modulus, so with half of the stress and (120/200) of the Young modulus, the strain for X will be 2 × (200/120) or times that of the strain for Y. The original lengths for X and Y are the same, so the extension of X will be 3.3 times that of Y.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
10			C	1	
			Total	1	
11			B	1	Examiner's Comments This question proved particularly straightforward and accessible to nearly all candidates.
			Total	1	
12			A	1	Examiner's Comments This question proved particularly straightforward and accessible to nearly all candidates.
			Total	1	
13			B	1	Examiner's Comments This question was slightly more challenging.
			Total	1	
14			B	1	
			Total	1	
15			C	1	
			Total	1	
16			A	1	
			Total	1	
17			D	1	
			Total	1	
18			C	1	
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
19		i	$E = \frac{1}{2}kx^2$ **or** E = mgh **or** 0.080 ×9.81 × 0.20 **or** $\frac{1}{2} \times 60 \times x^2$ 0.080 ×9.81 × 0.20 = $\frac{1}{2} \times 60 \times x^2$ *x* = 0.072 (m)	C1 C1 A1	
		ii	**1** Time of flight is independent of speed/AW Because distance of fall is the same **and** initial velocity vertically is zero / velocity is horizontal at **X** **2** *D* increases as speed at **X** increases because the time of flight is constant/AW *D* is directly proportional to speed at **X**	B1 B1 M1 A1	**Allow** algebraic answers that assume initial vertical velocity is zero/velocity is horizontal at X. **Allow** d = vt idea "*D* is directly proportional to speed at **X** because the time of flight is constant" scores 2. <u>Examiner's Comments</u> This part showed that many candidates thought that the time of flight of the car depended on the take-off speed of the car. Since the car is travelling horizontally the time of flight only depends on the height of the car above the horizontal track.
			Total	**7**	

Question			Answer/Indicative content	Marks	Guidance
20			**Level 3 (5–6 marks)** Clear description **and** clear analysis *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3–4 marks)** Some description **and** some analysis **or** Clear description **or** Clear analysis *There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.* **Level 1 (1–2 marks)** Limited description **or** Limited analysis *There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.* **0 marks** No response or no response worthy of credit.	B1× 6	Use level of response annotation in RM Assessor. **Indicative scientific points may include:** **Description** • Determine *T* by measuring several oscillations • Independent and dependent variables identified (e.g. *L* and *T*) • Variables kept constant (e.g. for L and T experiment, *m* is kept constant) • Repeating to determine average *T* • Measure length *L* and width *w* with ruler • Measure thickness t with a vernier (calliper) / micrometer • Use video/phone camera / stopwatch / data-logger and motion sensor / light gates and timer • Use top-pan balance / scales to measure *m* **Analysis** • Plot an <u>appropriate</u> graph, e.g. T^2 against L^3 **or** tabulate $T^2 \div L^3$ • Gradient of **best line** determined **or** average of $T^2 \div L^3$ • Use a large triangle to determine gradient • Gradient (or equivalent) related to E, e.g. gradient = $16\pi^2 m/wEt^3$ for T^2 against L^3 graph <u>Examiner's Comments</u> This part tested ideas about investigative experiments: there was a solid focus on elements of data-taking and instruments that should be used. Typically at A Level, analysis should include an appropriate graph and a comparison between the line of best fit and the equation under test. Putting the general equation below the given equation would make it much clearer how the candidate linked the gradient or y-intercept with the required property.
			Total	6	

Question			Answer/Indicative content	Marks	Guidance
21	a		Arrow vertical down <u>and</u> an arrow opposite to the frictional force. Both arrows labelled correctly.	M1 A1	 **Allow** weight / *mg* / W for the downward arrow <u>and</u> tension / *T* / 'force in rod' / 'force in tow bar' /'driving force' for the 'upward' arrow
	b		(W_s =) 1100 × 9.81 × sin 10° **or** 1100 × 9.81 × cos 80° (W_s = 1874 N or 1900 N)	C1 A0	**Allow** *g* instead of value
	c		force = 1900 + 300 force = 2200 (N)	A1	**Allow** 1870 + 300 = 2170 (N)
	d		(distance =) 120 / sin 10° **or** 691 (m) (work done =) 2200 × 691 work done = 1.5 × 10^6 (J)	C1 C1 A1	 **Allow** ECF from **(c)** **Allow** ECF from an incorrect attempt at first mark.
	e		(A =) π × 0.006^2 **or** 1.1 × 10^{-4} (m^2) (stress =) $\frac{2200}{\pi \times 0.006^2}$ and $2.0 \times 10^{11} = \frac{\text{stress}}{\text{strain}}$ x = 4.8 × 10^{-5} (m)	C1 C1 A1	**Allow** ECF from **(c)** **Allow** x (=FL/EA) = $\frac{2174 \times 0.5}{2.0 \times 10^{11} \times 1.1 \times 10^{-4}}$ **Allow** 2 marks for 1.2 × 10^{-5}; 1.2 × 10^{-2} m used as radius **Allow** answer between 4.7 and 5.1 × 10^{-5} (m)
			Total	**10**	

Question			Answer/Indicative content	Marks	Guidance
22	a		The **maximum** (tensile) **stress** a material can withstand (before it breaks)	B1	<u>Examiner's Comments</u> Ultimate tensile strength is the maximum stress a material can withstand without breaking or failing. The most common incorrect answer included descriptions of force rather than stress.
	b	i	Use a micrometer / (Vernier) caliper Measure the diameter along its length in different directions (to ensure uniform cross-section) AW	B1 B1	 **Allow** either along length or in different directions
		ii	**1.** Any value between 0 and 2.0 (kN) **2.** Any value between 2.0 and 2.2 (kN)	B1 B1	
		iii	k = gradient $k = 5.0 \times 10^5$ (N m^{-1})	C1 A1	 **Allow** 1 mark for 5.0×10^n; $n \neq 5$ **Allow** 5 x 105 (N m^{-1})
	c		(A =) $\pi \times 0.0021^2$ or 1.39×10^{-5} (m^2) (σ) = $2200 / \pi \times 0.0021^2$ $\sigma = 1.6 \pi 10^8$ (Pa)	C1 A1	 **Allow** 1 marks for $4(.0) \times 10^7$; diameter used as radius Answer is 1.59×10^8 Pa to 3sf

Question			Answer/Indicative content	Marks	Guidance
	d		Greater area under the graph (from 3 mm to 4 mm) / greater **average** force from (3 mm to 4 mm)	B1	ORA **Allow**: labelled/annotated diagram
			Mention of work done = **average** force x distance or work done = area under graph	B1	**Allow** energy (transferred) instead of work done **Allow** 2 marks for arguments including reference to $W = ½ kx^2$ **and** constant k **and** greater average x <u>Examiner's Comments</u> This question is best answered by referring to the graph in the question. Exemplars 1 and 2 indicate the difference between a low level and a high level response. **Exemplar 1** Work is the force aplied over a distance. If initial distance is larger then the work will be larger. [2] This exemplar shows a response that contains broadly true statements yet is only loosely linked to the context and so scored zero marks. **Exemplar 2** work done is equal to the area under the force extension graph and this area is larger for the area between 3 + 4 mm than 1 + 2 mm [2] This response is clearly and specifically about this context and uses the graph as supporting evidence. The link to the graph is that the area gives the work done and that a larger area for one region means a larger amount of work done for that region.
			Total	11	

Question			Answer/Indicative content	Marks	Guidance
23	a	i	Missing data point and error bar plotted correctly.	B1	**Allow** ½ square tolerance.
		ii	Force measured by pulling back plate with a newton–meter.	B1	
		ii	Extension measured with a ruler (placed close to the transparent plastic tube).	B1	
		iii	Best fit line drawn correctly and gradient determined correctly.	B1	**Ignore** POT for this mark; gradient = 50 ± 4 (N m^{-1})
		iii	Worst fit line drawn correctly and its gradient determined correctly.	B1	**Note**: The line must have a greater/smaller gradient than the best fit line and must pass through all the error bars. **Ignore** POT for this mark.
		iii	$2k = 50$ (N m^{-1}), therefore $k = 25$ (N m^{-1})	B1	Possible ECF.
		iii	Absolute uncertainty determined correctly.	B1	Possible ECF within calculation.
		iv	$F \propto x$ / straight line passing through the origin.	B1	
		v	energy stored = ½ × 50 × 0.12^2	C1	Possible ECF from **(iii)**
		v	½ × 50 × 0.12^2 = ½ × 0.39 × v^2	C1	**Allow** 1 mark for $v = 0.96$ m s^{-1}; used k for single spring
		v	$v = 1.4$ (m s^{-1})	A1	
	b		force constant of spring arrangement) = $\frac{2k}{3}$ $\frac{2k}{3}x = ma$	M1	
			$a = \frac{2}{3 \times 0.39}kx$	M1	
			$a = 1.7\ kx$	A0	
			Total	**13**	

Question			Answer/Indicative content	Marks	Guidance
24		i	(Sum of clockwise moments = sum of anticlockwise moments) 95 × 9.81 × 1.80 / 120 × 9.81 × 1.00 / 1.60 × *T* sin 30°	C1	
		i	(95 × 9.81 × 1.80) + (120 × 9.81 × 1.00) = 1.60 × *T* sin30°	C1	
		i	$T = 3.6 \times 10^3$ (N)	A1	**Note** answer to 3 s.f. is 3.57×10^3 (N)
		ii	$\sigma = \dfrac{3.6\times10^3}{\pi\times0.015^2}$	C1	Possible ECF from part **(i)**
		ii	$\sigma = 5.1 \times 10^3$ (kPa)	A1	**Allow** 1 mark for 5.1×10^6; POT error **Note** using 3.57×10^3 N gives 5.05×10^3 (kPa)
		iii	The clockwise moment increases and therefore *T* increases.	B1	
			Total	6	
25			The extension of each spring is halved because the force in each spring is halved. (Hence the force constant is *2k*.)	B1	**Allow** *F* = kx, *x* is halved for the same *F*, hence *k* doubles.
			Total	1	

Question			Answer/Indicative content	Marks	Guidance
26	a		*** Level 3 (5–6 marks)** All points E1, 2, 3 and 4 for equipment All points M1, 2, 3 and 4 for measurements For calculations expect C1, C2, C3 and C4 Expect at least two points from reliability *There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.* **Level 2 (3–4 marks)** Expect E1 and E2; E3 or E4 for equipment Expect M2 and two from M1, M3, M4 for measurements For calculations expect at least C3 and C4 Expect at least one point from reliability *There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.* **Level 1 (1–2 marks)** Expect at least E1 and E2 for equipment Expect at least two from measurements Expect C5 for the calculation No real ideas for obtaining reliable results *The information is basic and communicated in an unstructured way. The information is supported by limited evidence and the relationship to the evidence may not be clear.* **0 marks** No response or no response worthy of credit.	B1 × 6	The complete plan consists of four parts: **Equipment used safety (E)** 1. Wire fixed at one end with load added to wire. 2. Suitable scale with suitable marker on wire. 3. Micrometer screw-gauge **or** digital / vernier callipers for measuring diameter of wire. 4. Reference to safety concerning wire snapping. **Measurements (M)** 1. Original length from fixed end to marker on wire. 2. Diameter of wire. 3. Measure load. 4. New length of wire when load increased. **Calculation of Young modulus. (C)** 1. Find extension (for each load) or strain (for each load). 2. Determine cross-sectional area or stress. 3. Plot graph of load-extension **or** graph of stress-strain. 4. Young modulus = gradient × original length / area **or** Young modulus = gradient. 5. Calculate Young modulus from single set of measurements of load, extension, area and length. **Reliability of results (R)** 1. Measure diameter in 3 or more places and take average. 2. Put on initial load to tension wire and take up ‘slack’ before measuring original length. 3. Take measurements of extension while unloading to check elastic limit has not been exceeded. 4. Use long wire (to give measurable extension). Scale or ruler parallel to wire.

Question			Answer/Indicative content	Marks	Guidance
	b	i	Elastic: material returns to original dimensions when load is removed.	B1	
		i	Plastic: material has permanent change of shape when load is removed.		
		ii	The material is elastic because the removal of force returns the rubber to its original length.	B1	
		ii	The area under force-extension graph is work done.	B1	
		ii	Repeated stretching and releasing the rubber warms up the rubber because not all the strain energy is returned back. The area enclosed represents the amount of thermal energy. During landing, some of the aeroplane's kinetic energy is transferred to thermal energy and therefore the aeroplane does not "bounce" during landing; hence this minimises the risk to passengers.	B1	Mentioning 'hysteresis' is not enough to gain this mark.
			Total	10	